Selcom Payments Integration Guide

Checkout API + Mobile Push Payments + Card Payments + Status Checking + Qwiksend Payouts

Mwongozo huu umeandaliwa kwa kutumia taarifa kutoka Selcom Developers portal ili uwe na document moja ya utekelezaji kwa backend, frontend, webhook handler, na payout service.

Coverage Authentication, signing, create order, create-order-minimal, stored cards, wallet pull, Selcom Pesa push, webhook, order status, na Qwiksend payout.	**Audience** Backend developers, full-stack developers, SaaS/ecommerce teams, billing systems, marketplace systems, na subscription platforms.
Source date Portal ya Selcom Developers iliyopitiwa tarehe 17 Aprili 2026.	**Reference base** Authentication, API Response, Checkout API, na Qwiksend sections kutoka developers.selcommobile.com.

Kitu muhimu kabla ya ku-code

Selcom docs zinaonyesha endpoint family nyingi, lakini kwa biashara nyingi utahitaji core flows nne: create order, process payment, receive webhook, na verify status. Kwa payouts, flow kuu ni account lookup → bank transfer → query status.

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1. Muhtasari wa portal ya Selcom na SDKs

Portal ya Selcom Developers inaonesha API families kadhaa zikiwemo Utility Payments, Wallet Cashin, C2B/Collection Services, Qwiksend, VCN, Checkout API, Integrated Merchants, na IMT. Kwa matumizi ya ecommerce na billing, Checkout API ndiyo sehemu kuu ya card na mobile checkout, huku Qwiksend ikiwa sehemu ya bank disbursement / payout.

Eneo	Unalotumia kwa nini
Checkout API	Create order, hosted payment page, minimal non-card payment, stored cards, card charge, wallet pull, Selcom Pesa push, webhook, order status.
Qwiksend	Bank payout / disbursement kwenda bank account, plus account name lookup na status query.
C2B / Integrated Merchants	Inbound payment validation/notification upande wa merchant ERP.
VCN / IMT	Virtual cards na international transfer use cases; si lazima kwa checkout ya kawaida.

SDK na client packages zilizotajwa kwenye portal ni PHP, Java, C#, Python, Node.js, pamoja na WordPress na WooCommerce plugin options. Kwa backend mpya ya kisasa, Node.js au PHP ni njia rahisi zaidi kama tayari una server stack hizo.

2. Authentication, headers, na signing model

Selcom inahitaji API key na API secret. Kwa kila request unatuma set ya headers za usalama. Authentication imeelezwa kwa API key (base64 encoded ndani ya realm ya SELCOM), Timestamp ya ISO 8601, Digest-Method, Digest, na Signed-Fields.

Header	Maana ya kiutendaji
Authorization	Base64 encoded API key ukitangulizwa na realm ya SELCOM.
Timestamp	ISO 8601 datetime; hutumika pia kwenye signed string.
Digest-Method	HS256 au RS256 kulingana na setup ya merchant.
Digest	Base64 encoded signature ya timestamp + signed fields katika order ile ile iliyo kwenye Signed-Fields.
Signed-Fields	Majina ya fields zilizotiwa saini kwa mpangilio ule ule uliotumika kutengeneza Digest.

Signing formula ya msingi

Official docs zinaeleza kwamba Digest inatokana na data ya muundo kama: timestamp=<iso8601>&field1=value1&field2=value2 ... kwa mpangilio ule ule ulio kwenye Signed-Fields. Timestamp haisajiliwi kwenye Signed-Fields header lakini inashiriki kwenye signature.

```
Pseudo-sign input

timestamp=2026-04-17T12:00:00+03:00&vendor=SHOP001&order_id=INV-1001&amount=10000&currency=TZS

digest = Base64(HMAC_SHA256(secret, sign_input))
Authorization = "SELCOM " + Base64(api_key)
```

Tahadhari ya practical

Docs zinaonyesha mifano kadhaa yenye tofauti ndogo za field naming, kwa mfano buyer_user_id kwenye sample moja na buyer_userid kwenye nyingine, na no_of_items_items ikionekana kwenye Signed-Fields sample moja. Hivyo, test endpoint zako kwa sandbox au tumia official SDK/client package unapoweza badala ya ku-hardcode assumptions zote kwa mkono.

3. Common API response semantics

Kwa kawaida response ina reference, resultcode, result, message, na data. Result code 000 ni success. Result codes 111 au 927 zina maana ya transaction in progress, wakati 999 ni ambiguous; portal inapendekeza kufanya query ya status baada ya takriban dakika 3 ikiwa hali haijabadilika.

result / code	Tafsiri ya matumizi
SUCCESS / 000	Request imekamilika vizuri.
INPROGRESS / 111, 927	Muamala bado unaendelea; usitoe final success kwa user kabla ya query au webhook.
AMBIGOUS / 999	Status haijulikani bado; fanya retry ya status query na recon logic.
FAIL / nyingine zote	Request imefeli au upstream imekataa.

4. Checkout integration architecture ya kawaida

1. Frontend au backend huunda order mpya kwa kutumia create-order au create-order-minimal.
2. Selcom hurudisha metadata kama gateway_buyer_uuid, payment_token, qr, au payment_gateway_url.
3. Kwa hosted checkout, user anapelekwa payment gateway page. Kwa minimal checkout, merchant anaweza kuonyesha token/QR au kisha trigger wallet/card flow tofauti.
4. Baada ya malipo, Selcom inaweza kupiga webhook yako kwa successful transaction.
5. Backend yako inapaswa bado kufanya order-status verification kabla ya kuweka order kuwa paid/finalized.

Flow	Endpoint kuu	Inafaa kwa nini
Hosted full checkout	POST /v1/checkout/create-order	Card + wallet experience ya kawaida kupitia redirect page ya Selcom.
Minimal non-card checkout	POST /v1/checkout/create-order-minimal	Wallet/manual/QR use case ambapo hutaki card option ionekane.
Stored card direct charge	POST /v1/checkout/card-payment	Recurring, in-app, au stored-card flow bila redirect.
Wallet pull	POST /v1/checkout/wallet-payment	USSD push/pull authorization kwa mobile wallet ya mteja.
Selcom Pesa push	POST /v1/checkout/selcompesa-payment	Flow maalum ya Selcom Pesa push payment.

5. Hosted checkout: Create Order

Endpoint hii ndiyo standard ya kuanzisha order ya ecommerce kutoka merchant kwenda Selcom. Portal inaeleza kwamba card payment bila billing info inaweza kukataliwa, hivyo ukiwa unategemea cards hakikisha billing fields zako zipo au zimetumwa kama inavyotakiwa.

Field muhimu	Required?	Maelezo
vendor	Yes	Merchant / vendor ID uliyopewa na Selcom.
order_id	Yes	Unique order id ya merchant wako.
buyer_email	Yes	Email ya buyer.
buyer_name	Yes	Full name ya buyer.
buyer_phone	Yes	MSISDN ya buyer.
buyer_userid	Optional/Contextual	User id ya mfumo wako; muhimu kwa stored card retrieval.
gateway_buyer_uuid	Optional	Ikiwa buyer ameshawahi kuonekana kwenye gateway, tumia kurudisha stored cards.
amount	Yes	Order amount.
currency	Yes	TZS au USD kulingana na docs.
payment_methods	Yes	ALL, MASTERPASS, CARD, MOBILEMONEYPULL (comma-separated ikiwa zaidi ya moja).
redirect_url	Optional	Base64 encoded URL ya kurudisha buyer baada ya payment.
cancel_url	Optional	Base64 encoded URL ya kurudi buyer akicancel.
webhook	Optional	Base64 encoded callback URL ya merchant backend.

Response unachotakiwa kuhifadhi

Baada ya create-order, response inaweza kurudisha gateway_buyer_uuid, payment_token, qr, na payment_gateway_url. Hifadhi gateway_buyer_uuid kwa buyer wako; ndiyo ufunguo wa kupata stored cards baadaye.

```
Core idea

POST /v1/checkout/create-order
-> returns data[0].payment_gateway_url
-> frontend redirect user there
-> receive webhook on success
-> verify using GET /v1/checkout/order-status?order_id=...
```

6. Minimal checkout: Create Order Minimal

Hii ni version nyepesi ya create-order kwa non-card payments pekee. Official docs zinasema card option haitaonekana kwenye payment gateway page baada ya redirection. Ni nzuri kwa wallet push flows na case ambapo merchant anataka kuonyesha payment token au QR kwa mteja.

Tofauti na create-order ya kawaida	Create-order-minimal
Card support	Hapana; docs zinasema cannot be used for card payments.
Buyer fields	Buyer email, buyer name, buyer phone, amount, currency bado zinahitajika.
Best use case	Mobile wallet push, token display, QR display, offline/manual assisted checkout.
URL handling	Docs zinasema URLs kwenye request na response ni base64 encoded.

Wakati wa kuitumia

Tumia create-order-minimal ikiwa unaunda billing system, hotspot portal, kiosk, POS helper, au app ambayo inamuonyesha user token/QR bila full hosted card checkout.

7. Status lifecycle: cancel, order-status, list-orders, webhook

Hizi endpoints zinaunda lifecycle ya order baada ya creation. Usiegemee frontend redirect pekee kama final proof ya payment; tumia status verification na webhook pamoja.

Endpoint	Method	Matumizi
/v1/checkout/cancel-order	DELETE	Kufuta order kwa kutumia order_id kabla haijakamilika.
/v1/checkout/order-status	GET	Kuangalia hali ya order moja kwa order_id.
/v1/checkout/list-orders	GET	Kuleta orders kwa fromdate na todate; nzuri kwa reconciliation.
Merchant webhook URL	POST from Selcom	Kupokea taarifa ya successful payment kutoka gateway.

payment_status	Maana
PENDING	Payment haijafinalize bado.
COMPLETED	Payment imekamilika.
CANCELLED	Transaction/order imecancel.
USERCANCELLED / USERCANCELED	Buyer ameacha/cancel flow.
REJECTED	Authorization/payment imekataliwa.
INPROGRESS	Processing inaendelea, si final state.

Webhook rule ya busara

Official docs zinasema webhook ni kwa successful transactions. Hiyo ina maana usisubiri webhook pekee kujua kuwa malipo yamefeli; kwa failures, cancellations, au ambiguous states tumia order-status polling/retry policy.

6. Pokea webhook.
7. Validate signature / source kwa headers zile zile za Selcom.
8. Linganisha order_id na order yako ya ndani.
9. Piga order-status kwa order_id hiyo ili kuthibitisha final state.
10. Ikiwa payment_status ni COMPLETED ndipo uweke order kuwa paid, utoe huduma, au ufungue access.

8. Stored cards na direct card charging

Checkout API ina stored-card flow kwa recurring au in-app payments. Mteja akishawahi kupitia order creation na gateway_buyer_uuid kuhifadhiwa, unaweza baadaye ku-fetch card tokens zake na kucharge moja kwa moja bila redirect.

Step	Endpoint	Notes
Fetch stored cards	GET /v1/checkout/stored-cards	Inahitaji gateway_buyer_uuid na buyer_userid.
Delete stored card	DELETE /v1/checkout/delete-card	Inahitaji id ya card resource na gateway_buyer_uuid.
Charge stored card	POST /v1/checkout/card-payment	Inahitaji transid, vendor, order_id, card_token, buyer_userid, gateway_buyer_uuid.

Card token response field	Maana
masked_card	Card number iliyofichwa.
creation_date	Tarehe ya kuundwa kwa token.
name	Jina la card holder.
card_type	001 Visa, 002 Mastercard kulingana na docs.
card_token	Token ya kutumia kwa direct charge.
id	Resource id ya card record.

Direct card charge use case

Hii ndiyo flow ya subscription renewal, saved card checkout, au mobile app payment ambapo user hashitaji kuredirectiwa kila mara baada ya kwanza kuhifadhi card yake.

9. Mobile payment flows: Wallet Pull na Selcom Pesa Push

Portal ina flows mbili za mobile checkout ndani ya Checkout API ambazo zinafanana kwa payload structure ya msingi: transid, order_id, na msisdn.

Flow	Endpoint	Required fields	Use case
Wallet Pull Payment	POST /v1/checkout/wallet-payment	transid, order_id, msisdn	Push USSD / wallet authorization kutoka wallet ya billing customer.
Selcom Pesa Push Payment	POST /v1/checkout/selcompesa-payment	transid, order_id, msisdn	Selcom Pesa specific push payment flow.

Practical UX pattern

Kwanza create-order-minimal au create-order, kisha chagua flow ya wallet-payment au selcompesa-payment. Muonyeshe user “check phone to authorize” na usifungue huduma hadi webhook au order-status ithibitishe COMPLETED.

```
Suggested mobile sequence

1) POST /v1/checkout/create-order-minimal
2) POST /v1/checkout/wallet-payment  (or /selcompesa-payment)
3) Show waiting state to customer
```

```
4) Receive webhook or poll order-status
5) On COMPLETED -> grant service
```

10. Qwiksend payout / bank transfer

Qwiksend ndiyo bidhaa ya Selcom ya bank disbursement. Hapa ndipo unafanya payouts kutoka mfumo wako kwenda bank account ya beneficiary.

Hatua	Endpoint	Kwa nini ni muhimu
Account name lookup	GET /v1/qwiksend/lookup/	Hukagua jina la account kabla ya kutuma payout.
Bank transfer	POST /v1/qwiksend/process	Hutuma payout yenyewe kwenda bank account.
Query status	GET /v1/qwiksend/query	Huthibitisha final state ya payout kwa transid.

Bank transfer field	Required?	Maelezo
transid	Yes	Unique transaction id ya payout.
recipientFiCode	Yes	Bank shortcode ya destination bank.
recipientAccount	Yes	Account number ya anayepokea.
recipientName	Yes	Jina la beneficiary.
senderAccount	Yes	Source account au sender identifier kwenye platform yako.
senderName	Yes	Jina la anayelipa.
amount	Yes	Kiasi cha kutuma.
vendor	Yes	Float account identifier.
pin	Yes	Float account PIN.
msisdn	Yes	Namba ya simu ya sender.
purpose	Yes	Mfano GIFT; docs zinalitaja kama purpose ya transfer.
remarks	Optional	Description ya payout.

11. Lookup beneficiary account name kwa bank + account + transid.
12. Ikiwa jina limerudi sawa na matarajio, then tuma POST /v1/qwiksend/process.
13. Hifadhi transid, recipient account, amount, na request snapshot kwenye payouts table.
14. Query /v1/qwiksend/query kwa transid mpaka upate final answer au webhook/reconciliation process yako ihakikishe.

Payout best practice

Usitume payout moja kwa moja bila account lookup isipokuwa una sababu maalum ya biashara. Lookup hupunguza makosa ya beneficiary na disputes.

11. Recommended data model, webhook logic, na security checklist

Table	Fields muhimu za kuhifadhi
orders	order_id, user_id, amount, currency, status,

Table	Fields muhimu za kuhifadhi
	gateway_buyer_uuid, payment_token, selcom_reference, channel, phone, created_at, paid_at
payment_attempts	transid, order_id, method, msisdn/card_token, result, resultcode, request_payload, response_payload, verified_at
stored_cards	user_id, gateway_buyer_uuid, card_token, masked_card, card_type, created_at, is_active
webhook_logs	order_id, transid, reference, result, resultcode, payment_status, raw_headers, raw_payload, received_at
payouts	transid, beneficiary_bank, beneficiary_account, beneficiary_name, amount, status, resultcode, response_payload, processed_at

- Weka order_id na transid kuwa unique kwenye database yako.
- Treat webhook endpoint as idempotent: payload ikirudi mara mbili, isi-create duplicate fulfillment.
- Usi-trust frontend redirect kama proof of payment; trust webhook + status verification.
- Store raw request/response ya Selcom kwa audit, support, na reconciliation.
- Tumia retry policy kwa 111/927/999 states badala ya kuwaonyesha users instant fail au success.
- Sanitize na protect API key, API secret, vendor id, na float PIN kwenye environment variables pekee.
- Log signed string au hash metadata kwa debugging, lakini si vizuri kuhifadhi secrets wazi kwenye logs.

Production recommendation

Kwa biashara ya kweli, payment service iwe module tofauti yenye kazi hizi: signature builder, request sender, webhook verifier, status reconciler, na payout service. Hii hurahisisha maintenance na audit trail.

12. Implementation blueprint ya haraka

15. Omba credentials za Selcom: API key, API secret, vendor/till details, na float/pin details kama payout inahusika.
16. Jenga helper ya kuset headers zote tano: Authorization, Timestamp, Digest-Method, Digest, Signed-Fields.
17. Implement create-order flow na store gateway_buyer_uuid ukipata response.
18. Implement webhook endpoint na order-status verification endpoint kwenye backend yako.
19. Implement wallet/card flows kulingana na use case yako: hosted checkout, stored-card charge, au wallet push.
20. Implement Qwiksend payout module ikiwa mfumo wako unahitaji kutoa fedha kwenda bank.
21. Fanya sandbox/UAT cases: success, cancel, pending, duplicate webhook, duplicate order_id, duplicate transid, wrong signature, na timeout retry.

Appendix A: Endpoint quick sheet

Family	Method	Endpoint
Checkout	POST	/v1/checkout/create-order
Checkout	POST	/v1/checkout/create-order-minimal
Checkout	DELETE	/v1/checkout/cancel-order
Checkout	GET	/v1/checkout/order-status
Checkout	GET	/v1/checkout/list-orders
Checkout	GET	/v1/checkout/stored-cards
Checkout	DELETE	/v1/checkout/delete-card
Checkout	POST	/v1/checkout/card-payment
Checkout	POST	/v1/checkout/wallet-payment

Family	Method	Endpoint
Checkout	POST	/v1/checkout/selcompesa-payment
Qwiksend	GET	/v1/qwiksend/lookup/
Qwiksend	POST	/v1/qwiksend/process
Qwiksend	GET	/v1/qwiksend/query

Appendix B: Source notes

- Official source reviewed: Selcom Developers API Reference — https://developers.selcommobile.com/
- Sections used: Authentication, API Response, Qwiksend (Bank Transfer, Account Name Lookup, Query Transaction Status), Checkout API (Create Order, Create Order Minimal, Cancel Order, Get Order Status, List All Orders, Fetch Stored Card Tokens, Delete Stored Card, Process Order - Card Payment, Process Order - Wallet Pull Payment, Process Order - Selcom Pesa Push Payment, Webhook Callback).
- Guide compiled on: 17 Aprili 2026.

Mwisho

Hii guide imekusudiwa kukupeleka hadi hatua ya implementation. Ukiandika service layer zako vizuri, Selcom integration itakuwa set ya modules chache zenye responsibility wazi badala ya code iliyotawanyika kila sehemu ya mfumo.